SADIC: Methods Guide

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Outline

1. Clustering Preliminaries
2. Fuzzy Solid Clustering
3. Fuzzy Incomplete Data Clustering
4. Fuzzy Shell Clustering
5. Fuzzy Clustering Models
6. Global Quality Measures
7. Regression
8. Regression Quality Measures
9. Dimensionality Reduction
10. Supervised Classification

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Clustering versus classification

Classification

Supervised learning: objects are represented together with class labels, and the task is to infer a decision rule for unseen instances.

Clustering

Unsupervised learning: objects are unlabelled, and the task is to reveal hidden structure, similarity patterns or regularities in the data.

Key distinction: classification starts from known classes; clustering has to discover them.

Cluster analysis — cluster notions

- Cluster analysis groups objects so that members of the same cluster are, in some sense, more similar to each other than to objects in other clusters.
- Different notions of cluster are possible:
  - groups with small internal distances;
  - dense regions of the data space;
  - objects generated by the same probability distribution;
  - geometric configurations such as lines, ellipsoids or shells.
- Therefore cluster analysis is not one algorithm, but a family of models and methods.

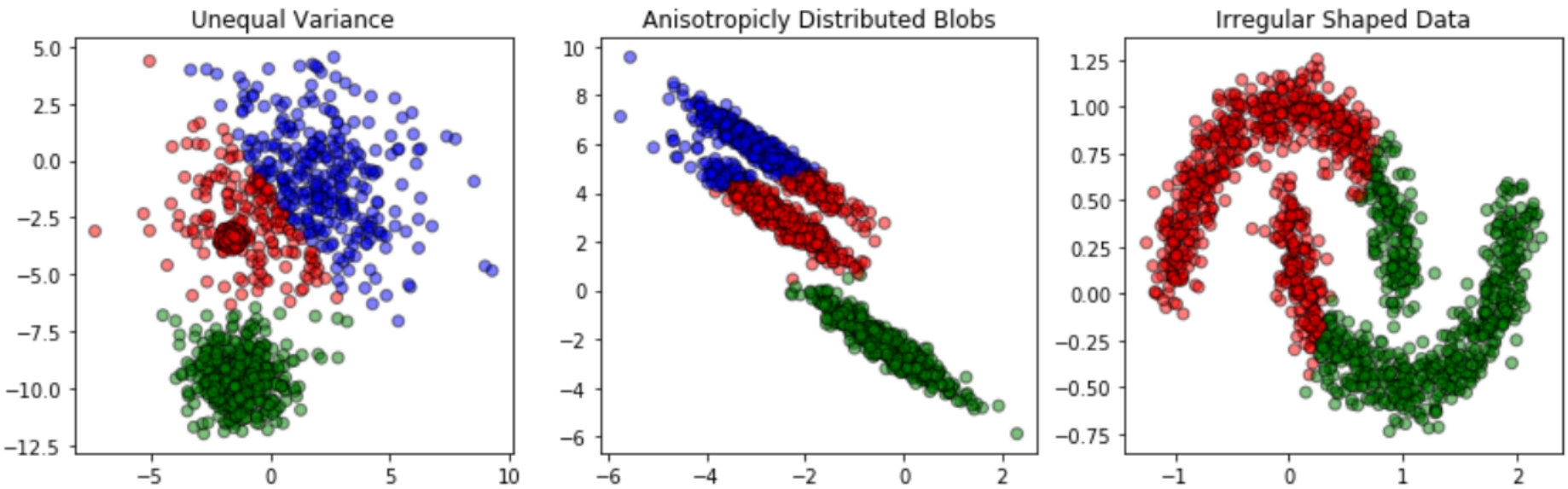
Clusterings and cluster models

- **Hard clustering:** an object belongs to one cluster or to none.
- **Fuzzy clustering:** an object belongs to each cluster to a certain degree.
- **Overlapping / hierarchical clustering:** objects may belong to several clusters or to nested clusters.
- Typical cluster models include:
  - connectivity models;
  - centroid models;
  - distribution models;
  - density models;
  - subspace / biclustering models;
  - shell or contour models.

No universally best clustering algorithm

- There is no objectively “correct” clustering algorithm.
- The appropriate method depends on the data, the expected cluster model and the intended use of the result.
- An algorithm designed for one model often fails on another:
  - *k*-means prefers compact, centroid-based clusters;
  - Gaussian mixtures are appropriate when a distribution model is plausible;
  - density-based methods detect arbitrary shapes, but depend on density separation;
  - shell methods are appropriate when objects concentrate near contours.
- Understanding the *cluster model* is therefore a prerequisite for understanding the algorithm.

Clustering issues



Different variances, anisotropy and irregular shapes motivate different clustering models.

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Fuzzy clustering – Discussion

Data representation Objects are vectors of measured values.

Feature extraction Reduce dimensionality and eliminate redundant characteristics.

Clusters shape Different geometric prototypes; norms or scalar products.

Clusters size Adaptive distances or adaptive algorithms.

Clusters validity Optimal number of classes through validity functionals, merging/splitting or hierarchical approaches.

Final fuzzy partition May need defuzzification; the original fuzzy partition should not be discarded.

Method Minimize a fuzzy objective function using a two-step iterative procedure that decreases its value.

Fuzzy clustering – Generic algorithm

Aim: minimize

$$J(P, L) = \sum_{i=1}^{c} \sum_{j=1}^{n} A_i(x^j)^m \, D(x^j, L_i).$$

1. Given c, n, m and x^j $(j = 1, \ldots, n)$; set $\ell = 0$.
2. Initialize $P^{(0)} = \{A_1, \ldots, A_c\}$.
3. Compute prototypes L_i minimizing $J(P^{(\ell)}, \cdot)$ (may be costly).
4. Compute $P^{(\ell+1)}$ minimizing $J(\cdot, L)$:

$$A_i^{(\ell+1)}(x^j) = \frac{1}{\sum_{k=1}^{c} \left(\frac{D(x^j, L_i)}{D(x^j, L_k)} \right)^{1/(m-1)}}.$$

5. If $P^{(\ell+1)}$ and $P^{(\ell)}$ are sufficiently close, stop; otherwise increment ℓ and repeat step 3.

Fuzzy clustering – Variants and improvements

Geometric prototypes

- Fuzzy c-Means – Dunn (1974), Bezdek (1974)
- Fuzzy c-Varieties and Fuzzy c-Elliptotypes – Bezdek et al. (1981)
- Fuzzy c-Ellipsoids – Lenart (1989)
- Adaptive Fuzzy Clustering – Dave (1989); Dumitrescu, Pop (1990)
- Fuzzy covariance matrix – Gustafson, Kessel (1979)
- L_p Fuzzy c-Means – Miyamoto, Agusta (1998); Hathaway, Bezdek (2000)

Fuzzy clustering – Variants and improvements: shell prototypes

Empty shell prototypes

- Fuzzy *c*-Shells – Dave (1990); Krishnapuram, Nasraoui, Frigui (1992)
- Adaptive Fuzzy *c*-Shells – Dave, Bhaswan (1992)
- Fuzzy *c*-Ellipsoidal Shells – Frigui, Krishnapuram (1996); Gath, Hoory (1995)
- Fuzzy *c*-Quadric Shells – Krishnapuram, Frigui, Nasraoui (1993, 1995)
- Fuzzy *c*-Rectangular Shells – Höppner, Klawonn, Kruse (1997)

Fuzzy clustering – Variants and improvements (2)

Fuzzy clustering with incomplete data

- Unsupervised fuzzy competitive learning – Chung, Lee (1994)
- Whole data strategy – Hathaway, Bezdek (2001)
- Partial distance strategy – Hathaway, Bezdek (2001)
- Optimal completion strategy – Hathaway, Bezdek (2001)
- Nearest neighbour strategy – Hathaway, Bezdek (2001)

Fuzzy clustering – Other methods and models

- Fuzzy divisive hierarchic clustering – Dumitrescu (1988)
- Cross-clustering – Dumitrescu, Pop (1995)
- Noise Clustering – Dave (1991); Dave, Sen (1997)
- Possibilistic, Probabilistic – Krishnapuram, Keller (1993); Gath, Geva (1989)

Fuzzy *C*-Means

- J.C. Dunn, *A Fuzzy Relative of the ISODATA Process and its Use in Detecting Compact, Well Separated Clusters*, Journal of Cybernetics 3 (1974), 32–54
- J.C. Bezdek, *Fuzzy Mathematics in Pattern Classification, Ph.D. Thesis, Applied Mathematics Center*, Cornell University, Ithaca, 1973

$$J_m^{FCM}(U, V; X) = \sum_{i=1}^{c} \sum_{k=1}^{n} u_{ik}^m d_{ik}^2$$

$$d_{ik} = \|x_k - v_i\|_A = \sqrt{(x_k - v_i)^T A (x_k - v_i)}$$

- A – symmetrical and positively defined

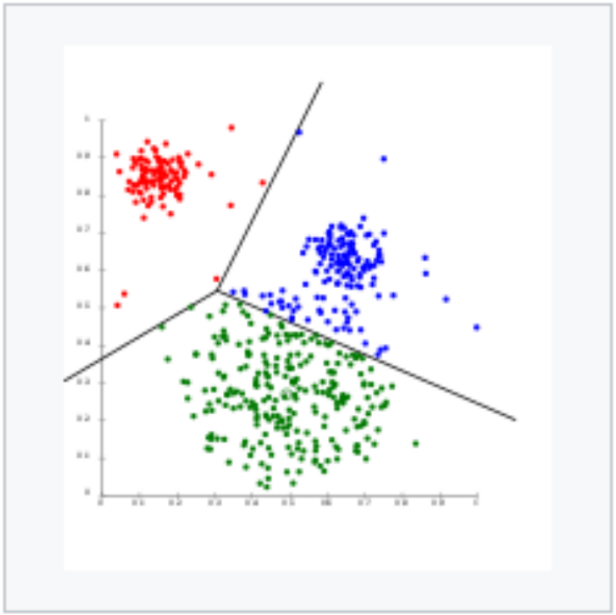
$$u_{ik} = \frac{1}{\sum_{j=1}^{c} \left(\frac{d_{ik}^2}{d_{jk}^2} \right)^{1/(m-1)}}$$

Fuzzy *C*-Means

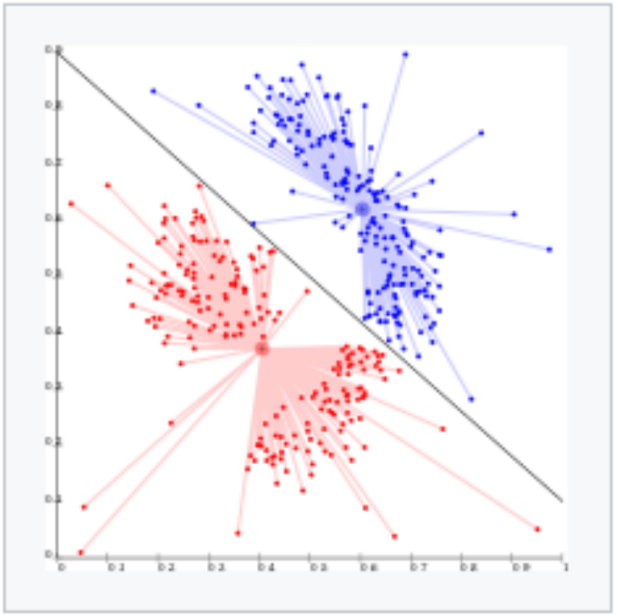
$$v_i = \frac{\sum_{k=1}^{n} u_{ik}^m x_k}{\sum_{k=1}^{n} u_{ik}^m}$$

1. initialize fuzzy partition
2. compute prototypes
3. compute new fuzzy partition
4. if fuzzy partitions are close enough, then stop, else goto step 2

Fuzzy *C*-Means — illustrative example



K-means separates data into Voronoi-cells, which assumes equal-sized clusters (not adequate here)



K-means cannot represent density-based clusters

Prototype-based methods induce Voronoi-like regions and work best for compact, centroid-oriented clusters.

Distance matrix A

1. $A = I$
2. $A = \text{Cov}^{-1}$

$$\text{Cov}_{ij} = \frac{1}{n-1}\sum_{k=1}^{n}(x_{ki} - \bar{x}_{.i}) \cdot (x_{kj} - \bar{x}_{.j})$$

3. $A = \text{Cor}^{-1}$

$$s_i^2 = \frac{1}{n-1}\sum_{k=1}^{n}(x_{ki} - \bar{x}_{.i})^2$$

$$\text{Cor}_{ij} = \frac{\text{Cov}_{ij}}{s_i \cdot s_j}$$

Similarity measure

$$T_A(x,y) = \frac{\langle x,y\rangle_A}{\|x\|_A^2 + \|y\|_A^2 - \langle x,y\rangle_A} = \frac{\langle x,y\rangle_A}{\|x-y\|_A^2 + \langle x,y\rangle_A}$$

- $T_A(x,y) \in [0,1]$, similarity measure
- $d_A(x,y) = 0 \iff T_A(x,y) = 1$
- $d_A(x,y) = \infty \Longrightarrow T_A(x,y) = 0$

Possibilistic Clustering

- R. Krishnapuram, J.M. Keller, *A Possibilistic Approach to Clustering*, IEEE Transactions on Fuzzy Systems 1, 2 (1993), 98–110

$$J_m^{PC}(U, P; X) = \sum_{i=1}^{c}\sum_{k=1}^{n} u_{ik}^m d_{ik}^2 + \sum_{i=1}^{c}\eta_i \sum_{k=1}^{n}(1-u_{ik})^m$$

- condition $\sum_{i=1}^{c} u_{ik} = 1$ is not required
- η_i is a sort of cluster radius: the square distance of the point with membership degree of 0.5 to class A_i

$$u_{ik} = \frac{1}{1 + \left(\frac{d_{ik}^2}{\eta_i}\right)^{1/(m-1)}}$$

$$\eta_i = \frac{\sum_{k=1}^{n} u_{ik}^m d_{ik}^2}{\sum_{k=1}^{n} u_{ik}^m}, \text{ if no } \textit{a priori} \text{ information is available}$$

Possibilistic Clustering

1. Run FCM
2. Compute η_i
3. Run PC

PCM — interpretation and limitations

- Each cluster defines its own *typicality function*; memberships are not constrained to sum to one.
- A point far from every prototype can have small typicality for all clusters.
- η_i controls the distance scale: $u_i(x) = 1/2$ when $d_i^2(x) = \eta_i$.
- Prototype and scale estimation are coupled. FCM initialization is frequently used to avoid poor local solutions.
- Limitation: two prototypes may converge to the same dense region (*coincident clusters*).

Interpretation: typicality is neither a posterior probability nor a normalized partition weight.

Noise Clustering

- R.N. Dave, *Characterization and Detection of Noise in Clustering*, Pattern Recognition Letters 12, 11 (1991), 657–664
- R.N. Dave, S. Sen, *On Generalizing the Noise Clustering Algorithms*, Proceedings of the 7-th IFSA World Congress, vol. III, Prague, 1997, 205–207

$$J_m^{NC}(U,P;X) = \sum_{i=1}^{c}\sum_{k=1}^{n} u_{ik}^m d_{ik}^2 + \sum_{k=1}^{n} \delta^2 \left(1 - \sum_{j=1}^{c} u_{jk}\right)^m$$

- virtual class with the “noise”
- hypothetical prototype $v^\star$, $d_{\star k} = \delta$
- condition $\sum_{i=1}^{c} u_{ik} < 1$ for all $k = 1, \ldots, n$

$$u_{ik} = \frac{1}{\sum_{j=1}^{c} \left(\frac{d_{ik}^2}{d_{jk}^2}\right)^{1/(m-1)} + \left(\frac{d_{ik}^2}{\delta^2}\right)^{1/(m-1)}}$$

Noise Clustering

$$\delta^2 = \frac{\lambda}{n \cdot c} \sum_{i=1}^{c} \sum_{k=1}^{n} d_{ik}^2 \quad \text{(Dave, 1991)}$$

- extension, δ_{ik} for each point and cluster

$$\delta_{ik} = \frac{\eta_i}{\sum_{j=1}^{c} \frac{d_{ik}^2}{d_{jk}^2}} \quad \text{(Dave, Sen, 1997)}$$

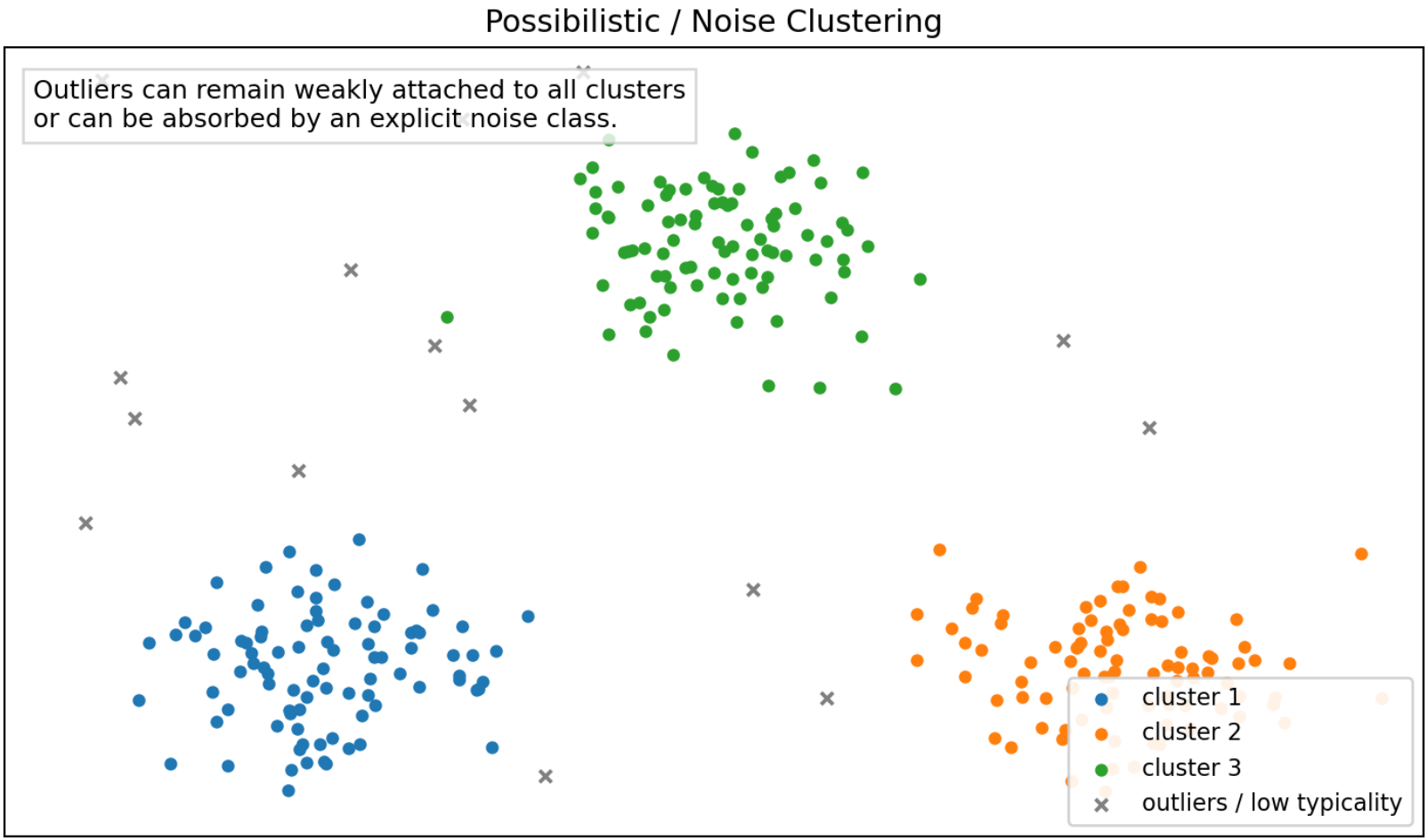
$$u_{ik}^{NC} = u_{ik}^{FCM} \cdot u_{ik}^{PCM}$$

Noise Clustering — rejection mechanism

- Add a virtual noise cluster at fixed distance δ from every object.
- Its membership equals $u_{0k} = 1 - \sum_{i=1}^{c} u_{ik}$.
- A distant object can be rejected by the real clusters without forcing it into one of them.
- The method retains competition between the real clusters: their degrees of membership remain coupled through normalization.
- The scale δ controls the strength of the rejection mechanism.

Contrast: Dave adds a rejection class; Krishnapuram removes partition normalization.

Possibilistic / Noise clustering — illustrative example



Low-typicality points need not be forced into ordinary clusters; they can remain weakly attached or be absorbed by a dedicated noise class.

Gustafson-Kessel Clustering

- E.E. Gustafson, W.C. Kessel, *Fuzzy Clustering with a Fuzzy Covariance Matrix*, IEEE CDC San Diego, CA (1979), 761–766

$$d_{ik} = \|x_k - v_i\|_{A_i};\ \det(A_i) = \rho_i,\ \text{input value}$$

- usually, $\rho_i = 1$
- v_i is computed as in FCM

$$S_i = \frac{\sum_{k=1}^{n} u_{ik}^m (x_k - v_i)(x_k - v_i)^T}{\sum_{k=1}^{n} u_{ik}^m}$$

$$A_i = (\rho_i \cdot \det(S_i))^{1/p} \cdot S_i^{-1}, p - \text{space dimension}$$

- u_{ik} is computed as in FCM

Gustafson–Kessel — geometric interpretation

- S_i estimates local scatter using fuzzy weights u_{ik}^m.
- A_i determines the orientation and axis ratios of the ellipsoidal distance contours.
- $\det(A_i) = \rho_i$ fixes the metric volume scale while allowing anisotropy.
- Memberships still use the FCM normalization: clusters remain mutually coupled.
- S_i must be invertible; near-singular matrices require regularization in implementations.

Main distinction: GK changes cluster geometry, not the interpretation of fuzzy membership.

Variant: diagonal matrix

$$A_i(k,k) = \frac{\sqrt[p]{\prod_{l=1}^{p}\sum_{j=1}^{n} u_{ij}^{m}(x_{jl} - v_{il})^2}}{\sum_{j=1}^{n} u_{ij}^{m}(x_{jk} - v_{ik})^2}$$

Gath-Geva Clustering

- I. Gath, A.B. Geva, *Unsupervised Optimal Fuzzy Clustering*, IEEE Trans. PAMI 11, 7 (1989), 773–781

$$d_{ik}^2 = d(x,(v,A,P))^2 = \frac{1}{P}\sqrt{\det(A)} \cdot \exp\left(\frac{1}{2}(x-v)^T A^{-1}(x-v)\right)$$

- N_i = normal probability distribution with expectation value v_i, covariance matrix A_i and *a-priori* probability P_i
- weighted component density $P_i f_i(x_k)$ (not an a-posteriori probability) is

$$P_i f_i(x_k) = \frac{1}{(2\pi)^{p/2}} \cdot \frac{1}{d_{ik}^2}$$

- v_i is computed as in FCM

$$A_i = \frac{\sum_{k=1}^{n} u_{ik}^m (x_k - v_i)(x_k - v_i)^T}{\sum_{k=1}^{n} u_{ik}^m}$$

Gath-Geva Clustering

$$P_i = \frac{\sum_{k=1}^{n} u_{ik}^m}{\sum_{k=1}^{n}\sum_{l=1}^{c} u_{lk}^m}$$

- u_{ik} is computed as in FCM

Gath–Geva — density and membership

For cluster i, the Gaussian density is

$$f_i(x) = \frac{\exp[-\frac{1}{2}(x - v_i)^T A_i^{-1}(x - v_i)]}{(2\pi)^{p/2}\sqrt{\det A_i}}.$$

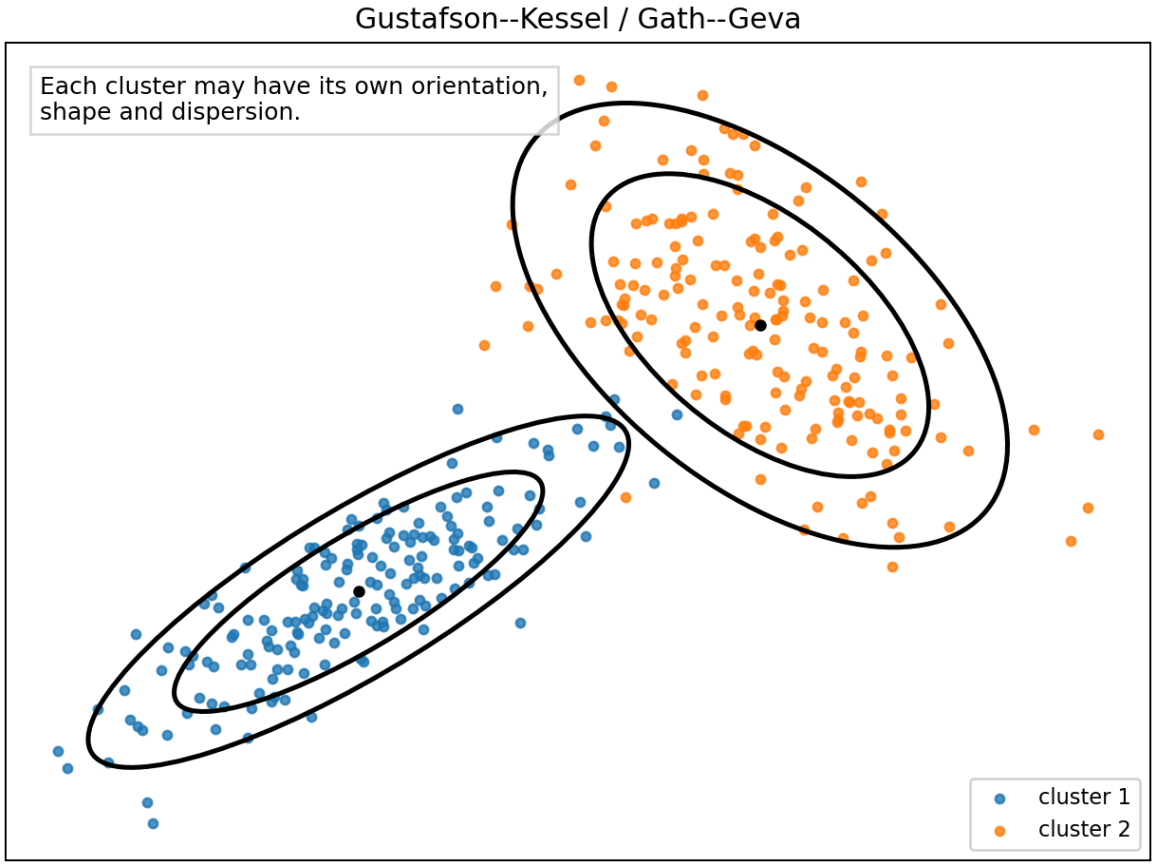
The GG distance satisfies $d_i^2(x) = (2\pi)^{-p/2}/[P_i f_i(x)]$.

- A_i adapts shape, orientation and volume; P_i adapts cluster weight.
- For $m = 2$, the FCM-type update has the same numerical form as normalized Gaussian mixture weights.
- Identical algebraic formulas need not imply identical semantics: GG optimizes a fuzzy partition; a mixture model infers a latent generating component.

Variant: diagonal matrix

$$A_i(k,k) = \frac{\sum_{j=1}^{n} u_{ij}^m (x_{jk} - v_{ik})^2}{\sum_{j=1}^{n} u_{ij}^m}$$

Adaptive metric clustering — illustrative example



Gustafson–Kessel and Gath–Geva adapt the metric (and, for Gath–Geva, also the density model) to clusters with different orientations and dispersions.

L_p Fuzzy C-Means

- S. Miyamoto, Y. Agusta, *Algorithms for L_1 and L_p Fuzzy C-Means and their Convergence*, Studies in Classification, Data Analysis, and Knowledge Organization: Data Science, Classification and Related Mehtods, C. Hayashi, N. Ohsumi, K. Yajima, Y. Tanaka, H.H. Bock, and Y. Baba, Eds., Tokyo, Japan, Springer Verlag, 1998, 295–302
- R.J. Hathaway, J.C. Bezdek, *Generalized Fuzzy C-Means Clustering Strategies Using L_p Norm Distances*, IEEE Transactions on Fuzzy Systems 8, 5 (2000), 576–582

$$
\begin{aligned}
J_{m,p}(U,v) &= \sum_{i=1}^{c}\sum_{k=1}^{n} u_{ik}^{m}\|x_k - v_i\|_p^p \\
&= \sum_{i=1}^{c}\sum_{k=1}^{n}\sum_{j=1}^{s} u_{ik}^{m}|x_{kj} - v_{ij}|^p
\end{aligned}
$$

L_p Fuzzy C-Means

- u_{ik} is computed as in FCM
- if $p > 1$, determine (by bisection method) v_{ij} as the root of

$$f'_{ij} = -p \sum_{k=1}^{n} u_{ik}^m |x_{kj} - v_{ij}|^{p-1} \text{sign}(x_{kj} - v_{ij}) = 0$$

- if $0 < p < 1$, v_{ij} is the smallest x_{kj} that globally minimizes

$$f_{ij} = \sum_{k=1}^{n} u_{ik}^m |x_{kj} - v_{ij}|^p$$

- if $p = 1$, v_{ij} is the mean of the smallest and largest x_{kj} that globally minimize f_{ij}

Fuzzy *C*-Varieties

- J.C. Bezdek, C. Coray, R. Gunderson, J. Watson, *Detection and Characterization of Cluster Substructure*, SIAM Journal of Applied Mathematics 40, 2 (1981), 339–372

$$d_{ik}^2 = d^2\left(x_k, (v_i, s_i)\right) = \|x_k - v_i\|_A^2 - \sum_{j=1}^{r} \left\langle x_k - v_i, s_{ij} \right\rangle_A^2$$

- s_{ij} – linear independent vectors
- r is size of variety, $r < p$
- v_i is computed as in FCM
- s_{ij} is the j-th eigenvector of covariance matrix S_i

$$S_i = A \cdot \frac{\sum_{k=1}^{n} u_{ik}^m (x_k - v_i)(x_k - v_i)^T}{\sum_{k=1}^{n} u_{ik}^m} \cdot A$$

- u_{ik} is computed as in FCM

Fuzzy *C*-Elliptotypes

- J.C. Bezdek, C. Coray, R. Gunderson, J. Watson, *Detection and Characterization of Cluster Substructure*, SIAM Journal of Applied Mathematics 40, 2 (1981), 339–372

$$J_m^{FCE} = (1-\alpha)J_m^{FCM} + \alpha J_m^{FCV}$$

$$d_{FCE}^2 = \|x-v\|_A^2 - \alpha \sum_{j=1}^{r} \langle x-v, s_j \rangle_A^2$$

Adaptive Fuzzy Clustering

- R.N. Dave, *Use of the Adaptive Fuzzy Clustering Algorithm To Detect Lines in Digital Images*, Proceedings Intelligent Robots and Computer Vision, VIII, vol. 1192 (1989), 600–611

$$d_{ik}^2 = d^2\left(x_k, (v_i, s_i); \alpha_i\right) = \|x_k - v_i\|_A^2 - \alpha_i \sum_{j=1}^{r} \left\langle x_k - v_i, s_{ij} \right\rangle_A^2$$

- $\lambda_{i,1}, \ldots, \lambda_{i,p}$ – eigenvalues of covariance matrix S_i

$$\alpha_i = 1 - \frac{\lambda_{i,p}}{\lambda_{i,1}} \quad \text{(Dave, 1989)}$$

$$\alpha_i = 1 - \frac{\dfrac{1}{p-r}\displaystyle\sum_{j=r+1}^{p} \lambda_{i,j}}{\dfrac{1}{r}\displaystyle\sum_{j=1}^{r} \lambda_{i,j}} \quad \text{(Bezdek, 1981)}$$

Variant: one weight per variety

- D.Dumitrescu, H.F. Pop, *Fuzzy Classification with Adaptive Prototypes*, Proceedings of the 2-nd Symposium on Fuzzy Partitions, Iaşi, 1990, 10–14

$$d_{ik}^2 = d^2\left(x_k, (v_i, s_i); \alpha_i\right) = \|x_k - v_i\|_A^2 - \sum_{j=1}^{r} \alpha_{i,j} \left\langle x_k - v_i, s_{ij} \right\rangle_A^2$$

- $\alpha_{i,j} = \bar{r}_{i,j} - \bar{r}_{i,j+1}$, where
- $\bar{r}_{i,j} = \dfrac{r_{i,j}}{r_{i,1}}$ are normalized radii
- $r_{i,1} \geq r_{i,2} \geq \cdots \geq r_{i,p}$ are computed as in Fuzzy C-Ellipsoids

Fuzzy *C*-Ellipsoids

- C. Lenart, *A Classification Algorithm for Ellipsoid Form Clusters*, University of Cluj-Napoca, Faculty of Mathematics Research Seminars on Computer Science, 9 (1989), 63–67

$$d_{ik}^2 = d^2\left(x_k, (v_i, s_i)\right) = \sum_{j=1}^{p} \frac{\left\langle x_k - v_i, s_{ij} \right\rangle_A^2}{r_{ij}^2}$$

- s_{ij} are linear independent vectors
- r_{ij} are ellipsoid radii
- v_i is computed as in FCM
- s_{ij} is computed as in FCV
- u_{ik} is computed as in FCM
- r_{ij} may be computed as follows

Fuzzy C-Ellipsoids

$$r_{ij} = \frac{\sum_{k=1}^{n} u_{ik}^{m} \left\langle x_k - v_i, s_{ij} \right\rangle_A^2}{\sum_{k=1}^{n} u_{ik}^{m}}, \text{ proportional to fuzzy variance along this direction}$$

$$r_{ij} = \max_{k=1,n} u_{ik} \left\langle x_k - v_i, s_{ij} \right\rangle_A, \text{ generalizes fuzzy radius}$$

$$r_{ij} = \lambda_{ij}, \text{ eigenvalues of the fuzzy covariance matrix}$$

1. Run FCM
2. Compute r_{ij}
3. Run FCE

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Unsupervised Fuzzy Competitive Learning

- F.L. Chung and T. Lee, *Fuzzy Competitive Learning*, Neural Networks 7, 3 (1994), 539–552
- the center v^i is determined via a competititve learning scheme

1. v_k^i initialized as random, for all i and k
2. iteration = 1; step = 1.0;
3. loop for all j, k: if x_k^j is available, then, for all i,

$$v_k^i = v_k^i + \text{step} \cdot (u_{ij})^m \cdot (x_k^j - v_k^i)$$

4. iteration $=$ iteration $+ 1$; step $=$ step$/$iteration
5. if old and new centers v are close enough, then stop, else goto step 3

Whole Data Strategy

- R.J. Hathaway, J.C. Bezdek, *Fuzzy C-Means Clustering of Incomplete Data*, IEEE Transactions on Systems, Man, Cybernetics, Part B: Cybernetics, 31, 5 (2001), 1062–1071
- delete incomplete data items
- membership of incomplete data items computed using the partial distance scheme (see below)

Partial Distance Strategy

- R.J. Hathaway, J.C. Bezdek, *Fuzzy C-Means Clustering of Incomplete Data*, IEEE Transactions on Systems, Man, Cybernetics, Part B: Cybernetics, 31, 5 (2001), 1062–1071
- consider only the present data components, i.e.

$$D_{ik} = \frac{s}{\sum_{j=1}^{s} I_{kj}} \cdot \sum_{j=1}^{s} \left(x_{kj} - v_{ij}\right)^2 I_{kj}$$

- $I_{kj} = 1$ if x_{kj} is present, and 0 if x_{kj} is missing
- the scaling factor in D_{ik} may be ignored

Optimal Completion Strategy

- R.J. Hathaway, J.C. Bezdek, *Fuzzy C-Means Clustering of Incomplete Data*, IEEE Transactions on Systems, Man, Cybernetics, Part B: Cybernetics, 31, 5 (2001), 1062–1071
- missing variables are optimised for minimisation of the objective function J_m
- randomly initialize missing variables
- at each step, after computing the membership degrees and the prototypes, update the missing variables based on

$$x_{kj} = \frac{\sum_{i=1}^{c} (u_{ik})^m \cdot v_{ij}}{\sum_{i=1}^{c} (u_{ik})^m}$$

Nearest Neighbour Strategy

- R.J. Hathaway, J.C. Bezdek, *Fuzzy C-Means Clustering of Incomplete Data*, IEEE Transactions on Systems, Man, Cybernetics, Part B: Cybernetics, 31, 5 (2001), 1062–1071
- similar with the Optimal Completion Strategy
- the missing variables are updated based on

$$x_{kj} = v_{ij}, \text{ where } D_{ik} = \min\{D_{1k}, \ldots, D_{ck}\}$$

Incomplete data — four strategies

- **Whole data**: estimate clusters from complete cases; assign incomplete cases afterwards.
- **Partial distance**: compare only observed coordinates, with optional rescaling.
- **Optimal completion**: estimate missing values while minimizing the FCM objective.
- **Nearest neighbour**: fill missing values from the closest prototype.

Important: deleting missing observations, ignoring missing coordinates, and estimating missing values solve different problems.

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Fuzzy *C*-Shells

- R.N. Dave, *Fuzzy Shell Clustering and Application to Circle Detection in Digital Images*, International Journal on General Systems 16 (1990), 343–355

$$d^2(x,(v,r)) = (\|x-v\|_A - r)^2$$

- optimal (v,r) verifies

$$\sum_{k=1}^{n} u_{ik}^m \left(1 - \frac{r_i}{\|x_k - v_i\|_A}\right)(x_k - v_i) = 0$$

$$\sum_{k=1}^{n} u_{ik}^m \left(\|x_k - v_i\|_A - r_i\right) = 0$$

Fuzzy *C*-Shells

1. $$v_i = \frac{\sum_{k=1}^{n} u_{ik}^m \cdot x_k}{\sum_{k=1}^{n} u_{ik}^m}$$
2. $$r_i = \frac{\sum_{k=1}^{n} u_{ik}^m \cdot \|x_k - v_i\|_A}{\sum_{k=1}^{n} u_{ik}^m}$$
3. $$v_i(next) = \frac{\sum_{k=1}^{n} u_{ik}^m \cdot \left(1 - \frac{r_i}{\|x_k - v_i\|_A}\right) \cdot x_k}{\sum_{k=1}^{n} u_{ik}^m \cdot \left(1 - \frac{r_i}{\|x_k - v_i\|_A}\right)}$$
4. goto step 2 or stop

Fuzzy C-Spherical Shells

- R. Krishnapuram, O. Nasraoui, H. Frigui, *The Fuzzy C-Spherical Shells Algorithm: A New Approach*, IEEE Transactions on Neural Networks 3 (1992), 663–671

$$d^2(x,(v,r)) = \left(\|x-v\|^2 - r^2\right)^2, \qquad y_k = (x_{k1} \ \ldots \ x_{kp} \ 1)^T$$

$$w_i = \sum_{k=1}^{n} u_{ik}^m \|x_k\|^2 y_k, \qquad H_i = \sum_{k=1}^{n} u_{ik}^m y_k y_k^T,$$

$$q_i = H_i^{-1} w_i, \qquad v_i = \frac{1}{2}(q_{i1} \ \ldots \ q_{ip})^T,$$

$$r_i^2 = \|v_i\|^2 + q_{i,p+1}.$$

Adaptive Fuzzy *C*-Shells

- R.N. Dave, K. Bhaswan, *Adaptive Fuzzy C-Shell Clustering and Detection of Ellipses*, IEEE Transactions on Neural Networks 3 (1992), 643–662

$$d^2(x,(v,r)) = (\|x-v\|_A - r)^2$$

- determine v_i and r_i as in FCS
- determine A_i as in Gustafson-Kessel
- determine u_{ik} as in FCS

Fuzzy C-Ellipsoidal Shells

- H. Frigui, R. Krishnapuram, *A Comparison of Fuzzy Shell Clustering Methods For Detection of Ellipses*, IEEE Transactions on Fuzzy Systems 4 (1996), 193–199

$$d^2(x,(v,A)) = \left(\frac{\|x-v\|_A - 1}{\|x-v\|_A}\right)^2 \cdot \|x-v\|^2$$

- optimal (v,r) verifies

$$0 = \sum_{k=1}^{n} u_{ik}^m \frac{\sqrt{d_{ik}}-1}{d_{ik}^2}\left(\|x_k - v_i\|^2 A_i + (\sqrt{d_{ik}}-1)d_{ik} I\right)(x_k - v_i)$$

$$0 = \sum_{k=1}^{n} u_{ik}^m \frac{\sqrt{d_{ik}}-1}{d_{ik}^2}(x_k - v_i)(x_k - v_i)^T \|x_k - v_i\|^2$$

where $d_{ik} = \|x_k - v_i\|_{A_i}^2$

Fuzzy C-Ellipses

- I. Gath, D. Hoory, *Fuzzy Clustering of Elliptic Ring-Shaped Clusters*, Pattern Recognition Letters 16 (1995), 727–741

$$d^2(x,(v_0,v_1,r)) = \left(\|x-v_0\|_A + \|x-v_1\|_A - r\right)^2, \quad x \in \mathbb{R}^2$$

- optimal (v_0, v_1, r) verifies

$$0 = \sum_{k=1}^{n} u_{ik}^m \left((x_k - v_{i,l}) + (\|x_k - v_{i,1-l}\|_A - r_i) \frac{(x_k - v_{i,l})}{\|x_k - v_{i,l}\|_A} \right)$$

$$0 = \sum_{k=1}^{n} u_{ik}^m \left(\|x_k - v_{i,0}\|_A + \|x_k - v_{i,1}\|_A - r_i \right)$$

Fuzzy *C*-Ellipses

1. $v_i = \dfrac{\sum_{k=1}^{n} u_{ik}^m \cdot x_k}{\sum_{k=1}^{n} u_{ik}^m}$
2. $v_{i,0} = v_i + f_i \cdot e_{i,0}$, $v_{i,1} = v_i - f_i \cdot e_{i,0}$, where
   - $f_i = \dfrac{\sqrt{\lambda_{i,0} - \lambda_{i,1}}}{2}$
   - $e_{i,0}$, $e_{i,1}$ and $\lambda_{i,0}$, $\lambda_{i,1}$ are the eigenvectors and eigenvalues of the fuzzy covariance matrix
     $\sum_{k=1}^{n} u_{ik}^m \cdot \left(\|x_k - v_{i,0}\|_A + \|x_k - v_{i,1}\|_A \right)$
3. $r_i = \dfrac{}{\sum_{k=1}^{n} u_{ik}^m}$
4. $v_{i,l}(next) = \dfrac{\sum_{k=1}^{n} u_{ik}^m \left(x_k + (\|x_k - v_{i,1-l}\|_A - r_i) \dfrac{(x_k - v_{i,l})}{\|x_k - v_{i,l}\|_A} \right)}{\sum_{k=1}^{n} u_{ik}^m}$
5. goto step 3 or stop

Fuzzy *C*-Quadric Shells

- R. Krishnapuram, H. Frigui, O. Nasraoui, *The Fuzzy C-Quadric Shell Clustering Algorithm and the Detection of Second-Degree Curves*, Pattern Recognition Letters 14 (1993), 545–552.
- R. Krishnapuram, H. Frigui, O. Nasraoui, *Fuzzy and Possibilistic Shell Clustering Algorithms and their Applications to Boundary Detection and Surface Approximation*, IEEE Transactions on Fuzzy Systems 3 (1995), 29–60.

- $X \subset \mathbb{R}^p$, $r = p(p+1)/2$; prototype: a quadric polynomial in $\mathbb{R}^p$.
- For $p = 2$:

$$a_1 x_1^2 + a_2 x_2^2 + a_3 x_1 x_2 + a_4 x_1 + a_5 x_2 + a_6.$$

- The algebraic squared residual is

$$d^2(x, a) = \left(x^T \Phi(a) x + x^T (a_{r+1}, \ldots, a_{r+p})^T + a_{r+p+1}\right)^2.$$

The prototype describes a boundary, not a cluster centre.

Fuzzy C-Quadric Shells — coefficients

$$\Phi(a) \in \mathbb{R}^{p\times p}, \quad \Phi_{ij}(a) = \begin{cases} a_i & i = j, \\ \frac{1}{2} a_{ip-i(i-1)/2+j-1} & i < j, \\ \Phi_{ji}(a) & i > j. \end{cases}$$

$$p_i = (a_{i,1}, \ldots, a_{i,p}, a_{i,p+1}/\sqrt{2}, \ldots, a_{i,r}/\sqrt{2})^T,$$
$$q_i = (a_{i,r+1}, \ldots, a_{i,r+p+1})^T.$$

$$r_k = (x_{k1}^2, \ldots, x_{kp}^2, \sqrt{2}x_{k1}x_{k2}, \ldots, \sqrt{2}x_{k,p-1}x_{kp})^T,$$
$$s_k = (x_{k1}, \ldots, x_{kp}, 1)^T.$$

$$R_i = \sum_{k=1}^{n} u_{ik}^m r_k r_k^T, \qquad S_i = \sum_{k=1}^{n} u_{ik}^m s_k s_k^T.$$

Fuzzy *C*-Polynomials

H.F. Pop (2001). Polynomial prototype and weighted least-squares estimation:

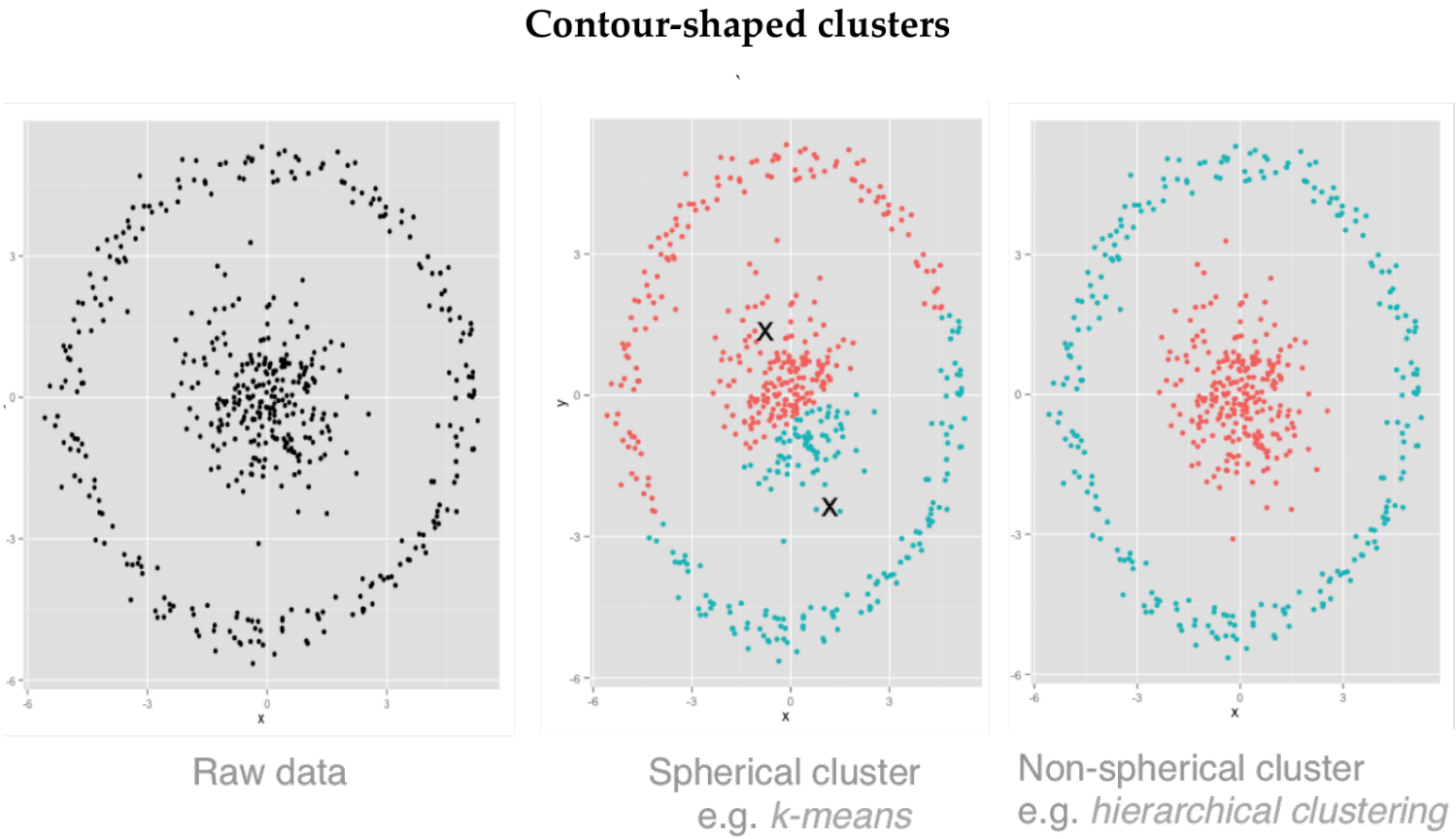
$$y = a_0 + a_1 x + \cdots + a_p x^p, \qquad d^2((x,y),a) = \left(\sum_{j=0}^{p} a_j x^j - y \right)^2.$$

$$s_i^{(j)} = \sum_{k=1}^{n} u_{ik}^m x_k^j, \qquad t_i^{(j)} = \sum_{k=1}^{n} u_{ik}^m x_k^j y_k.$$

$$S_{i,jk} = s_i^{(j+k-2)}, \quad j,k = 1,\ldots,p+1.$$

$$(a_{i,0},\ldots,a_{i,p})^T = S_i^{-1}(t_i^{(0)},\ldots,t_i^{(p)})^T.$$

Shell clustering — illustrative example



Shell models are appropriate when the relevant information lies on contours rather than around centroids.

Solid clusters versus shell clusters

- **Solid model**: closeness to a centre or an interior prototype.
- **Shell model**: closeness to a boundary, such as a circle, sphere, ellipse or quadric.
- A point at a circle's centre is not close to its shell, even though it is close to the centre.
- Choosing the prototype family determines which geometric structures can be detected.

Example: circle detection minimizes residuals such as $(\|x - v\| - r)^2$, not $\|x - v\|^2$.

Shell fitting — numerical considerations

- Shell clustering fits geometric parameters using fuzzy observation weights.
- Circular shells require centres and radii; ellipses additionally require shape and orientation.
- Quadric models are expressive, but their algebraic residuals depend on coefficient scaling; impose normalization or suitable constraints.
- Initialization, noise and ill-conditioned parameter estimates can affect convergence.

Connection: weighted polynomial fitting is closely related to regression within each class.

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Fuzzy Divisive Hierarchic Clustering

- D. Dumitrescu, *Hierarchical Pattern Classification*, Fuzzy Sets and Systems 28 (1988), 145–162
- binary fuzzy classification hierarchy
- binary partition index, $R(P)$ larger than a given threshold
- $P = \{C_1, C_2\}$, fuzzy partition of fuzzy set C

$$R(P) = \frac{\sum_{i=1}^{2} \sum_{k=1}^{n} C_{i,1/2}(x_k)}{\sum_{k=1}^{n} C(x_k)}$$

$$C_{i,t}(x) = \begin{cases} C_i(x) & C_i(x) > t \\ 0 & \text{otherwise} \end{cases}$$

Simultaneous (Cross) Fuzzy *C*-Means

- D. Dumitrescu, H.F. Pop, C. Sârbu, *Fuzzy Hierarchical Cross-Classification of Greek Muds*, Journal of Chemical Information and Computer Sciences 35 (1995), 851–857

1. Run FCM on data set $X \in \mathbb{R}^{n \times p}$
2. Run FCM on data set $\bar{Y} \in \mathbb{R}^{p \times c}$

$$\bar{y}_{ki} = \sum_{j=1}^{n} a_{ij} \cdot x_{jk}, \quad k = 1, \ldots, p; \; i = 1, \ldots, c$$

3. Run FCM on data set $\bar{X} \in \mathbb{R}^{n \times c}$

$$\bar{x}_{ji} = \sum_{k=1}^{p} b_{ik} \cdot x_{jk}, \quad j = 1, \ldots, n; \; i = 1, \ldots, c$$

4. If partitions are close enough, then stop, else go to step 2

$\Rightarrow$ Final grouping: (A_i, B_i), $i = 1, \ldots, c$.

Associative Simultaneous (Cross) Fuzzy C-Means

- H.F. Pop, C. Sârbu, *The Fuzzy Hierarchical Cross-Clustering Algorithm. Improvements and Comparative Study*, Journal of Chemical Information and Computer Sciences 37 (1997), 510–516

1. Run SFCM
2. Compute matrix

$$z_{k,l} = \sum_{j=1}^{n} \sum_{i=1}^{p} a_{kj} \cdot b_{li} \cdot x_{ij}, \quad k, l = 1, \ldots, c$$

3. Determine permutation σ that maximizes the function

$$J(\sigma) = \prod_{i=1}^{c} z_{i,\sigma(i)}$$

4. Relabel fuzzy sets B_i so that $B_{\sigma(i)}$ becomes B_i, for all i.

$\Rightarrow$ Final grouping: (A_i, B_i), $i = 1, \ldots, c$.

Fuzzy Hierarchic Simultaneous (Cross) Clustering

- H.F. Pop, C. Sârbu, *The Fuzzy Hierarchical Cross-Clustering Algorithm. Improvements and Comparative Study*, Journal of Chemical Information and Computer Sciences 37 (1997), 510–516

Variant A

- use ASFCM at each node of the binary hierarchy

Variant B

- at all nodes, except the root node, instead of starting ASFCM with original data X, use modified data $\bar{X}$ as last produced when clustering the parent node

Variant C

- at each node use ASFCM, except that instead of using modified data $\bar{X}$, always use original data X

Cross-clustering — interpretation

- Ordinary clustering groups observations using their features.
- Cross-clustering alternates between partitions of observations and partitions of features.
- The two partitions inform one another through aggregated data representations.
- Label alignment is required to establish which observation group corresponds to which feature group.

Output: associated pairs of fuzzy classes, one on each axis of the data matrix.

Biclustering

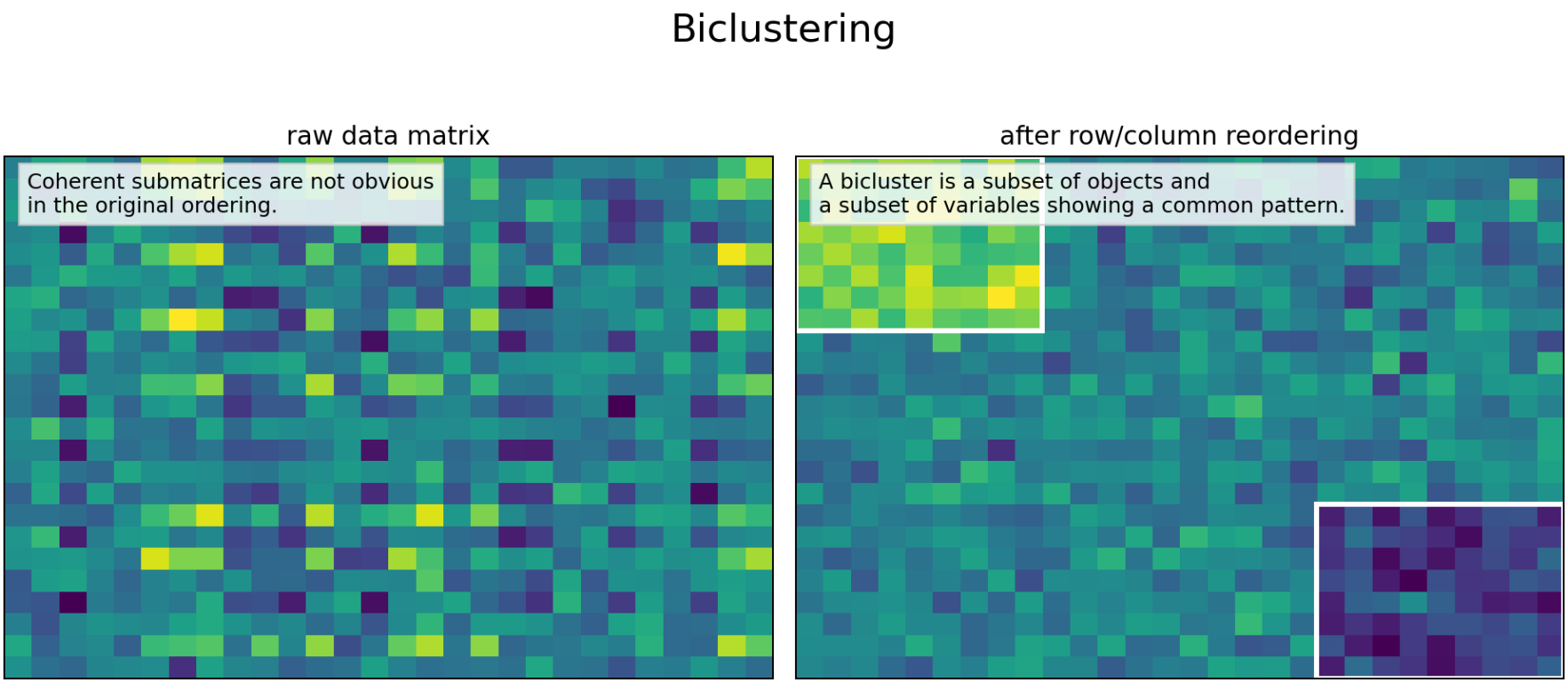
- Clustering is one of the most popular approaches in data analysis.
- Traditional clustering partitions data according to similarity across all variables.
- Some items behave similarly only on a subset of variables; their behavior may be uncorrelated on the other variables.
- Biclustering methods address this situation.
- In practice, bicluster boundaries are often fuzzy.
- Fuzzy biclusters provide richer information than crisp biclusters.

Fuzzy *C*-Bipartitioning

Biclustering algorithms: select both items and variables so that the chosen items are similar on the chosen variables. **Our aim:** generalize this approach.

- Produce fuzzy biclusters, not crisp ones.
- Produce a set of biclusters, not only one.
- A fuzzy bicluster is composed of two fuzzy sets:
  - **Variables:** weights in the distance used to construct the fuzzy set of items.
  - **Items:** weights in the distance used to construct the fuzzy set of variables.
- Iterative procedure comprising two calls to fuzzy clustering algorithms with adaptive metrics.

Biclustering — illustrative example



A bicluster is a subset of objects and a subset of variables that exhibit a common local pattern.

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Cluster validity — what is being measured?

- **Partition-only measures** (PC, PE, Backer–Jain): evaluate the distribution of memberships, without inspecting the data geometry.
- **Geometry-based measures** (Xie–Beni, separation): evaluate compactness and separation using distances or prototypes.
- **Volume/density measures** (FHV, APD, PD): use cluster covariance and membership-weighted observations.
- These criteria are not interchangeable: a very crisp partition need not identify well-separated clusters.

Partition Coefficient

- J.C. Bezdek, *Cluster Validity with Fuzzy Sets*, Journal of Cybernetics 3, 3 (1974), 58–72

$$PC = \frac{1}{n} \sum_{k=1}^{n} \sum_{i=1}^{c} u_{ik}^2$$

- to maximize
- $PC = 1/c \iff$ uniform fuzzy partition
- $PC = 1 \iff$ crisp partition

Normalized Partition Coefficient

$$NPC = \frac{PC \cdot c - 1}{c - 1}$$

- to maximize
- $NPC = 0 \iff$ uniform fuzzy partition
- $NPC = 1 \iff$ crisp partition

Partition Entropy

- J.C. Bezdek, *Mathematical Models for Systematics and Taxonomy*, Proceedings of the 8-th International Conference on Numeric Taxonomy, San Francisco, CA, 1975, 143–166

$$PE = -\frac{1}{n}\sum_{k=1}^{n}\sum_{i=1}^{c} u_{ik} \ln u_{ik}$$

- to minimize
- $PE = 0 \iff$ crisp partition
- $PE = \ln c \iff$ uniform fuzzy partition

Normalized Partition Entropy

$$NPE = \frac{PE}{\ln c}$$

- to minimize
- $NPE = 0 \iff$ crisp partition
- $NPE = 1 \iff$ uniform fuzzy partition

Proportion Exponent

- M.P. Windham, *Cluster Validity for Fuzzy Clustering Algorithms*, Fuzzy Sets and Systems 3 (1980), 1–9

$$PX = -\ln \prod_{k=1}^{n} \left(\sum_{j=1}^{[1/\mu_k]} (-1)^{j+1} \binom{c}{j} (1 - j\mu_k)^{c-1} \right)$$

$$\mu_k = \max_{i=1,c} u_{ik}$$

- to maximize
- $PX = 0 \iff$ uniform fuzzy partition
- $PX = \infty \iff$ crisp partition

Backer-Jain Index

- E. Backer, A.K. Jain, *A Clustering Performance Measure Based on Fuzzy Set Decomposition*, IEEE Transactions on Pattern Analysis and Machine Intelligence, PAMI-3, 1 (1981), 66–74

$$IBJ = \frac{1}{n(c-1)} \sum_{i=1}^{c-1} \sum_{j=i+1}^{c} \sum_{k=1}^{n} |u_{ik} - u_{jk}|$$

- to maximize
- $IBJ = 0 \iff$ uniform fuzzy partition
- $IBJ = 1 \iff$ crisp partition

Xie-Beni Index

- X.L. Xie, G.A. Beni, *Validity Measure for Fuzzy Clustering*, IEEE Transactions on Pattern Analysis and Machine Intelligence, PAMI-13, 8 (1991), 841–846

$$IXB = \frac{1}{n} \frac{\sum_{i=1}^{c} \sum_{k=1}^{n} u_{ik}^2 \|x_k - v_i\|^2}{\min_{i \neq j} \|v_i - v_j\|^2}$$

- to minimize

Separation Coefficient

- R. Gunderson, *Application of Fuzzy ISODATA Algorithms to Star Tracker Pointing Systems*, Proceedings of the 7-th Triennial World IFAC Congress, Helsinki, 1978, 1319–1323

$$SC = 1 - \max_{i<j} \frac{r_i + r_j}{\|v_i - v_j\|}$$

$$r_i = \max\{\|x - v_i\| | x \in A_i\}$$

- crisp partition
- to minimize
- $0 < SC < 1 \iff$ no pair of classes intersect one another
- $SC = 0 \iff$ the closest pair of classes are tangent
- $SC < 0 \iff$ at least one pair of classes intersect one another

Separation Index

- J.C. Dunn, *Well Separated Clusters and Optimal Fuzzy Partitions*, Journal of Cybernetics 4-1 (1974), 95–104

$$D_1 = \frac{\min\limits_{i \neq j = 1,c} d(A_i, A_j)}{\max\limits_{k=1,c} \text{diam}(A_k)}$$

$$D_2 = \frac{\min\limits_{i \neq j = 1,c} d(A_i, \text{conv}(A_j))}{\max\limits_{k=1,c} \text{diam}(A_k)}$$

$$d(A_i, A_j) = \min\{d(x, y) | x \in A_i, y \in A_j\}$$

Separation Index

$$\text{diam}(A_k) = \max\{d(x, y)|x, y \in A_k\}$$

- crisp partition
- to maximize
- $D_1 > 1 \iff$ partition formed by c compact, separated (CS) clusters (definition)
- $D_2 > 1 \iff$ partition formed by c compact, well-separated (CWS) clusters (definition)
- $D_1 \geq D_2 \geq D_1 - 1$

Separation

$$S = \frac{\sum_{i=1}^{c}\sum_{k=1}^{n} u_{ik}^2 \|x_k - v_i\|^2}{c \cdot \min_{k,l} d^2(A_k, A_l)}$$

- to minimize
- $S \leq \frac{1}{D_1^2}$

Fuzzy Hypervolume

- I. Gath, A.B. Geva, *Unsupervised Optimal Fuzzy Clustering*, IEEE Trans. PAMI 11, 7 (1989), 773–781

$$FHV = \sum_{i=1}^{c} \sqrt{\det(F_i)}$$

$$F_i = \frac{\sum_{k=1}^{n} u_{ik}^2 (x_k - v_i)(x_k - v_i)^T}{\sum_{k=1}^{n} u_{ik}^2}$$

- to minimize

Average Partition Density

- I. Gath, A.B. Geva, *Unsupervised Optimal Fuzzy Clustering*, IEEE Trans. PAMI 11, 7 (1989), 773–781

$$APD = \frac{1}{c}\sum_{i=1}^{c}\frac{S_i}{\sqrt{\det(F_i)}}$$

$$S_i = \sum_{j\in Y_i} u_{ij}, \quad Y_i = \{k \leq n | \|x_k - v_i\|_{F_i^{-1}} < 1\}$$

- to maximize

Partition Density

- I. Gath, A.B. Geva, *Unsupervised Optimal Fuzzy Clustering*, IEEE Trans. PAMI 11, 7 (1989), 773–781

$$PD = \frac{\sum_{i=1}^{c} S_i}{FHV}$$

- to maximize

Shell Clustering Measures

- $\Delta_{ik} = x_k - z_{i,k}$, for $k = 1, \ldots, n$, $i = 1, \ldots, c$
- $z_{i,k}$ is the point on the shell of class A_i which is the closest to x_k

FHV, APD, PD

- R.N. Dave, *New Measures for Evaluating Fuzzy Partitions Induced Through C-Shell Clustering*, Proceedings SPIE Conference Intell. Robot. Computer Vision, X, vol. 1607 (1991), 406–414
- Fuzzy Shell Covariance Matrix

$$A_i = \frac{\sum_{k=1}^{n} u_{ik}^m \Delta_{ik} \Delta_{ik}^T}{\sum_{k=1}^{n} u_{ik}^m}$$

Shell Thickness

- R. Krishnapuram, H. Frigui, O. Nasraoui, *Fuzzy and Possibilistic Shell Clustering Algorithms and their Applications to Boundary Detection and Surface Approximation*, IEEE Transactions on Fuzzy Systems 3 (1995), 29–60

$$T_S = \sum_{i=1}^{c} \frac{\sum_{k=1}^{n} u_{ik}^m \|\Delta_{ik}\|^2}{\sum_{k=1}^{n} u_{ik}^m}$$

- for $A_i(v_i, r_i)$, Average Circle Thickness

$$T_C = \frac{T_S}{\frac{1}{c}\sum_{i=1}^{c} r_i}$$

$$\|\Delta_{ik}\|^2 = (\|x_k - v_i\| - r_i)^2$$

Validity for shell clustering

- A solid-cluster index measures dispersion around a centre; this can misrepresent a ring or curved shell.
- For shell models, residuals from the nearest point on the fitted surface are more informative.
- Shell thickness measures weighted deviations *normal to the shell*, not distances to its centre.
- Interpret any index in the context of the fitted prototype family and the chosen number of classes.

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Regression techniques – The problem

- Aim: model a measured response from explanatory variables
- Common approach: linear least squares
- Assumption: homoscedasticity; error variance independent of predictor
- Heteroscedastic observations and outliers motivate robust methods
- No single robust method is optimal in all situations
- Quality measures can depend on the method
- Fuzzy weights for individual observations; validate comparatively and by cross-validation

Fuzzy regression – Discussion

- Clustering reveals substructure, but ordinary fuzzy partition methods are unsuitable for $c = 1$
- Find one fuzzy set A and one prototype L describing the data
- Use a second, hypothetical class $\bar{A}$ with constant dissimilarity

$$D(x^j, \bar{L}) = \delta = \left(\frac{\alpha}{1 - \alpha} \right)^{m-1}$$

$$J(A, L) = \sum_{j=1}^{n} A(x^j)^m D(x^j, L) + \delta \sum_{j=1}^{n} \bar{A}(x^j)^m, \quad \alpha \in (0, 1)$$

Fuzzy regression – Generic algorithm

1. Given α; initialize $A^{(0)}(x) = 1$, $l = 0$.
2. Compute prototype L minimizing $J(A^{(l)}, \cdot)$.
3. Compute $A^{(l+1)}$ minimizing $J(\cdot, L)$:

$$A^{(l+1)}(x^j) = \frac{\alpha/(1-\alpha)}{\alpha/(1-\alpha) + D(x^j, L)^{1/(m-1)}}.$$

4. Stop when successive fuzzy sets are sufficiently close; otherwise iterate.

Improvement: use relative distances for scale independence.

Fuzzy regression – Properties

- $A(x) = 1$ if and only if $D(x, L) = 0$.
- If $D_r(x, L) = 1$, then $A(x) = \alpha$ (relative-distance formulation).
- Membership decreases as dissimilarity increases.
- The parameter α controls the polarization of $\{A, \bar{A}\}$.
- Linear prototype: weighted centroid and principal fuzzy covariance direction.
- Use orthogonal rather than vertical distances when fitting geometric lines.

Fuzzy Regression

- H.F. Pop, C. Sârbu, *A New Fuzzy Regression Algorithm*, Analytical Chemistry 68 (1996), 771–778

$$J_m^{FR}(U, P; X) = \sum_{k=1}^{n} u_k^m d_k^2 + \sum_{k=1}^{n} \delta (1 - u_k)^m$$

- only one real class, $d_k = d(x_k, \mathbf{p})$
- a second, virtual class
- hypothetical prototype $\mathbf{p}^\star$, $d_{\star k}^2 = \delta$

$$u_k = \frac{1}{1 + \left(\frac{d_k^2}{\delta} \right)^{1/(m-1)}}$$

$$\delta = \lambda \cdot \max_{k=1,n} d_k^2$$

- $\lambda = \frac{\alpha}{1 - \alpha}$, where α is the given membership degree assigned to the farthest point from the prototype

Robust regression – Other approaches

- Non-fuzzy methods: weighted least squares, iteratively reweighted least squares
- Weights may be based on estimated variance or residuals
- Fuzzy regression: fuzzy observations, fuzzy coefficients, or fuzzy weights
- Other formulations: minimum fuzziness, error least squares, interval analysis
- Chang and Ayyub (2001): alternative fuzzy regression approaches

Linear, Single — Least Squares

- Linear Fit: $y = a + bx$; Quadratic Fit: $y = a + bx + cx^2$; Polynomial Fit: $y = a + bx + cx^2 + \ldots$

$$J^{LS}(x, y, a) = \sum_{k=1}^{n} \left(a_0 + a_1 x_k + \cdots + a_p x_k^p - y_k\right)^2$$

$$s^{(i)} = \frac{1}{n} \sum_{k=1}^{n} x_k^i, \qquad r^{(i)} = \frac{1}{n} \sum_{k=1}^{n} x_k^i y_k$$

$$S_{ij} = s^{(i+j-2)}, \quad R_i = r^{(i-1)}, \qquad i, j = 1, \ldots, p+1$$

$$\boxed{a = S^{-1} R}$$

Linear, Multiple — Least Squares

- Linear Fit: $y = a_0 + a_1 x_1 + \cdots + a_p x_p$.

$$J^{LS}(x, y, a) = \sum_{k=1}^{n} \left(a_0 + \sum_{j=1}^{p} a_j x_{kj} - y_k \right)^2$$

$$s^{(i,j)} = \frac{1}{n} \sum_{k=1}^{n} x_{ki} x_{kj}, \quad r^{(i)} = \frac{1}{n} \sum_{k=1}^{n} x_{ki} y_k$$

By convention $x_{k0} = 1$, so $s^{(0,j)}, s^{(i,0)}, s^{(0,0)}$ and $r^{(0)}$ are defined.

$$S_{ij} = s^{(i-1,j-1)}, \quad R_i = r^{(i-1)}, \qquad i, j = 1, \ldots, p+1$$

$$\boxed{a = S^{-1} R}$$

Nonlinear — Levenberg–Marquardt

Model $y = f(x; a)$.

$$J^{LM}(x, y; a) = \sum_{k=1}^{n} (y_k - f(x_k; a))^2$$

$$G_i = \sum_{k=1}^{n} (y_k - f(x_k; a)) \frac{\partial f(x_k; a)}{\partial a_i}$$

$$C_{ij} = \sum_{k=1}^{n} \frac{\partial f(x_k; a)}{\partial a_i} \frac{\partial f(x_k; a)}{\partial a_j}$$

$$M_{ij} = \begin{cases} (1 + \lambda) C_{ii} & \text{if } i = j, \\ C_{ij} & \text{if } i \neq j. \end{cases}$$

Levenberg–Marquardt — algorithm

1. Initialize a and evaluate $J^{LM}(x, y; a)$.
2. Initialize $\lambda = 0.001$ (conservatively).
3. Compute $\delta a = M^{-1} G$ and evaluate $J^{LM}(x, y; a + \delta a)$.
4. If the objective does not decrease, multiply λ by 10 and retry step 3.
5. Otherwise accept $a \leftarrow a + \delta a$ and continue iteration.
6. Stop when the objective change is smaller than ε.

Regression models — Exponential Family

- Exponential Fit, $y = a \exp(bx)$
- Modified Exponential Fit, $y = a \exp(b/x)$
- Logarithm Fit, $y = a + b \ln x$
- Reciprocal Logarithm Fit, $y = 1/(a + b \ln x)$
- Vapor Pressure Model, $y = \exp(a + b/x + c \ln x)$

Regression models — Power Law Family

- Power Fit, $y = ax^b$; Modified Power, $y = ab^x$
- Shifted Power, $y = a(x - b)^c$; Geometric, $y = ax^{bx}$
- Modified Geometric, $y = ax^{b/x}$; Root Fit, $y = a^{1/x}$
- Hoerl Model, $y = ab^x x^c$; Modified Hoerl Model, $y = ab^{1/x} x^c$

Regression models — Yield-Density and Growth

Yield-Density Models

- Reciprocal, $y = 1/(a + bx)$; Reciprocal Quadratic, $y = 1/(a + bx + cx^2)$
- Bleasdale, $y = (a + bx)^{-1/c}$; Harris, $y = 1/(a + bx^c)$

Growth Models

- Exponential Association (2), $y = a(1 - \exp(-bx))$
- Exponential Association (3), $y = a(b - \exp(-cx))$
- Saturation Growth, $y = ax/(b + x)$

Regression models — Sigmoidal Models

- Gompertz, $y = a\exp(-\exp(b - cx))$
- Logistic, $y = a/(1 + \exp(b - cx))$
- Richards, $y = a/(1 + \exp(b - cx))^{1/d}$
- MMF, $y = (ab + cx^d)/(b + x^d)$
- Weibull, $y = a - b\exp(-cx^d)$

Regression models — Miscellaneous Models

- Sinusoidal Fit, $y = a + b\cos(cx + d)$
- Gaussian Model, $y = a\exp(-(x-b)^2/(2c^2))$
- Hyperbolic Fit, $y = a + b/x$
- Heat-Capacity Model, $y = a + bx + c/x^2$
- Rational Function, $y = (a + bx)/(1 + cx + dx^2)$

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Regression Quality Measures

$$r_k = y_k - \hat{y}_k, \qquad \bar{o} = \frac{1}{n}\sum_{k=1}^{n} o_k$$

$$QC_1 = \sqrt{\frac{1}{n-1}\sum_{k=1}^{n}\left(\frac{r_k}{\hat{y}_k}\right)^2}$$

$$QC_2 = \sqrt{\frac{1}{n-1}\sum_{k=1}^{n}\left(\frac{r_k}{y_k}\right)^2}$$

$$QC_3 = \sqrt{\frac{1}{n-1}\sum_{k=1}^{n}\left(\frac{r_k}{\bar{\hat{y}}}\right)^2}$$

$$QC_4 = \sqrt{\frac{1}{n-1}\sum_{k=1}^{n}\left(\frac{r_k}{\bar{y}}\right)^2}$$

Regression Quality Measures (continued)

$$QC_5 = \sqrt{\sum_{k=1}^{n} \left(\frac{r_k}{\max_{i=1,\ldots,n} |r_i|} \right)^2}, \qquad NC_5 = \frac{QC_5}{\sqrt{n}} \in [0, 1]$$

$$QC_6 = \sqrt{\sum_{k=1}^{n} \left(\frac{r_k}{\bar{r}} \right)^2}, \qquad NC_6 = \frac{QC_6}{n} \in [0, 1]$$

$$ESR = \sqrt{\frac{1}{n-2} \sum_{k=1}^{n} r_k^2}$$

- QC_5 and QC_6: Pop and Sârbu (1996).
- For nonzero residuals: $QC_5 \in [1, \sqrt{n}]$, $QC_6 \in [\sqrt{n}, n]$; for all zero residuals both are defined as zero.

Regression quality — interpretation and safeguards

- The quality criteria in the guide normalize residuals in different ways; compare only measures having the same definition.
- A denominator containing y_k, $\hat{y}_k$ or a residual mean can vanish: the associated score then needs a stated convention or is undefined.
- The usual residual standard error divides by residual degrees of freedom; $n - 2$ is appropriate for a two-parameter linear model, not universally.
- Always inspect signed residuals as well as any aggregate measure: a low average error can conceal systematic structure.

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PCA — Overview

- Reduce dimensionality by replacing original variables with fewer principal components.
- Components follow directions of maximum variance and are linear combinations of the original variables.
- Choose a retained variance threshold or another component-count criterion.
- Isolated points can distort estimated principal directions.

PCA — Algorithm

1. Compute the covariance (or correlation) matrix:

$$\mathrm{Cov}_{ij} = \frac{1}{n-1} \sum_k (x_{ki} - \bar{x}_i)(x_{kj} - \bar{x}_j)$$

$$s_i^2 = \mathrm{Cov}_{ii}, \quad \mathrm{Cor}_{ij} = \frac{\mathrm{Cov}_{ij}}{s_i s_j}$$

2. Compute eigenvectors and eigenvalues; select retained components.
3. Project centered data onto selected eigenvectors.

PCA — Coordinate Transformation

$$Y = A^T \cdot X$$

- $A \in \mathbb{R}^{s \times s}$ is the matrix of column eigenvectors of covariance matrix
- $X \in \mathbb{R}^{s \times n}$ is the matrix of column data vectors
- $Y \in \mathbb{R}^{s \times n}$ is the matrix of column transformed vectors

Fuzzy PCA, first component

- Down-weight isolated observations according to their distance to the first fuzzy principal direction.
- Fit a linear prototype using fuzzy regression.

$$C_{ij} = \frac{\sum_{k=1}^{n} A(x^k)^m (x_{ki} - \bar{x}_i)(x_{kj} - \bar{x}_j)}{\sum_{k=1}^{n} A(x^k)^m}$$

- The leading eigenvector of C is the first fuzzy principal component.
- This does not, on its own, robustify all remaining components.

Fuzzy PCA, orthogonal

- Compute the first fuzzy principal direction.
- Project the data onto the orthogonal complement of that direction.
- Recompute the first fuzzy direction in the reduced-dimensional subspace.
- Repeat recursively, then map the eigenvectors back into the original coordinates.
- Advantage: robustifying additional components reduces to successive first-component problems.

FuzzyPCAFirst Algorithm

1. Select α by optimizing a quality criterion for the first component.
2. Determine fuzzy weights for this α.
3. Calculate the corresponding fuzzy covariance matrix C.
4. Compute its eigenvectors and eigenvalues: the fuzzy components and scatter measures.

FuzzyPCAOrthogonal Algorithm

1. If $p \leq 2$, use FuzzyPCAFirst; otherwise continue.
2. Optimize α, determine fuzzy weights and covariance matrix.
3. Take the leading eigenvector e with maximal eigenvalue λ.
4. Project data into the orthogonal complement of e.
5. Recursively compute the fuzzy PCA of reduced-dimensional data.
6. Express the recovered eigenvectors in the original coordinates.

Dimension of Reduced Space

Value of eigenvalues. Keep only factors with eigenvalue higher than 1.

Difference of eigenvalues. Keep factors until the difference of consecutive eigenvalues is smaller than a given threshold.

Sum of eigenvalues. Chose the number of preserved factors so that the ratio of the sum of largest eigenvalues to the sum of all eigenvalues is higher than a given threshold.

Variable Selection or Projection

Data projection. Set $d < s$, and select d eigenvectors as above: $A \in \mathbb{R}^{s \times d}$ and $Y \in \mathbb{R}^{d \times n}$.

Minimal angles. For each considered principal component, select the original variable that forms the minimal angle with it.

Highest loading. For each considered principal component, select the original variable that corresponds to the highest loading; if already selected, select the following.

Communalities. Select the variables with highest communalities.

$$c_i = \sum_{j=1}^{d} A_{ij}^2, \quad i = 1, \ldots, s$$

Multi-Dimensional Scaling

$$d : \mathbb{R}^{p \times p} \to \mathbb{R}, \text{ pseudo-metric} \implies d' : \mathbb{R}^{s \times s} \to \mathbb{R}, \text{ metric}$$

- Construct a low-dimensional configuration that preserves pairwise (dis)similarities.
- Metric and non-metric formulations preserve different information.

Metric MDS

- R.N. Shepard, *The Analysis of Proximities: Multidimensional Scaling with an Unknown Distance Function*, Psychometrika 27, 2 (1962), 125–140

$$E_M = \sum_{i \neq j} \left(d_{ij} - d'_{ij} \right)^2$$

$$E_M(\text{normalized}) = \frac{\sum_{i \neq j} \left(d_{ij} - d'_{ij} \right)^2}{\sum_{i \neq j} \left(d_{ij} \right)^2}$$

Non-metric MDS

- J.B. Kruskal, *Nonmetric Multidimensional Scaling: A Numerical Method*, Psychometrika 29 (1964), 115–129
- f is monotonically increasing function to map distances so that it preserves rank order

$$E_N = \frac{\sum_{i \neq j} \left(f(d_{ij}) - d'_{ij} \right)^2}{\sum_{i \neq j} \left(d'_{ij} \right)^2}$$

Sammon mapping

- J.W. Sammon, *A Nonlinear Mapping for Data Analysis*, IEEE Transactions on Computers, C-18 (1969), 401–409
- preserves local distances

$$E_S = \frac{\sum_{i \neq j} \frac{\left(d_{ij} - d'_{ij}\right)^2}{d_{ij}}}{\sum_{i \neq j} d_{ij}}$$

PCA versus multidimensional scaling

- **PCA**: produces orthogonal directions maximizing projected variance; requires vector-valued data.
- **Metric MDS**: finds coordinates that reproduce given dissimilarities as accurately as possible.
- **Non-metric MDS**: focuses on preserving the ordering of dissimilarities.
- **Sammon mapping**: gives relatively high weight to small original dissimilarities.

Caution: an informative low-dimensional display does not by itself establish that the original clusters are preserved.

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(Fuzzy) Perceptron

- F. Rosenblatt, *The Perceptron: A Probabilistic Model for Information Storage and Organization in the Brain*, Psychological Review 65 (1958), 386–408
- $X = \{x_1, \ldots, x_n\} \subset \mathbb{R}^p$; $\{A_1, A_2\}$ is a (fuzzy) partition of X

$$y_k = \begin{pmatrix} x_j \\ 1 \end{pmatrix} \in \mathbb{R}^{p+1}$$

$$z_k = \begin{cases} y_k & \text{if } a_{1k} > 0.5 \\ -y_k & \text{if } a_{2k} > 0.5 \end{cases}$$

- determine $v \in \mathbb{R}^{p+1}$ so that $v^T z_k > 0$ for all k

(Fuzzy) Perceptron

1. Chose $c > 0$ and initialize $v \in \mathbb{R}^{p+1}$ arbitrarily and $k = 1$
2. Update v

$$v(new) = \begin{cases} v + c \cdot a_{ik} \cdot z_k & \text{if } a_{ik} > 0.5 \text{ and } v^T z_k \leq 0 \\ v & \text{otherwise} \end{cases}$$

3. If v has not been modified during last n iterations, then stop, else increase k and goto step 2

(Fuzzy) Gallant Pocket Algorithm

- S.I. Gallant, *Perceptron-based Learning Algorithms*, IEEE Transactions on Neural Networks 1, 2 (1990), 179–191
- memorize in w the best v so far
- run(v) = number of consecutive correct classifications
- num(v) = total number of correct classifications

Finite training set

1. Initialize $k = 1$ and run(v) = run(w) = num(v) = num(w) = 0
2. Randomly select training vector z
3. If v correctly classifies z, then
   1. Increase run(v) and k
   2. If run(v) > run(w) then
      1. Compute num(v)
      2. If num(v) $>$ num(w), then assign v to w and update run(w) and num(w)
      3. If num(w) = n, then stop
4. Else, update v and initialize run(v) to 0 and increase k

- start with $i = 10000$ iterations
- if w has been updated during the last $0.8i$ iterations, then increase $i = 1.5i$ and continue with i more iterations

Infinite training set

1. Initialize $k = 1$ and run(v) = run(w) = 0
2. Randomly select training vector z
3. If v correctly classifies z, then
   1. Increase run(v) and k
   2. If run(v) $>$ run(w), then assign v to w and update run(w)
4. Else, update v and initialize run(v) to 0 and increase k

Training rules

- the first r training vectors are rules that must not be failed

1. Initialize $k = 1$ and run(v) = run(w) = num(v) = num(w) = 0
2. If v does not classify correctly the rules, select the training vector z as one of the rules; otherwise, randomly select training vector z from the rest of the data
3. If v correctly classifies z, then
    1. Increase run(v) and k
    2. If run(v) > run(w) then
        1. Compute num(v)
        2. If num(v) $>$ num(w), then assign v to w and update run(w) and num(w)
        3. If num(w) = n, then stop
4. Else, update v and initialize run(v) to 0 and increase k

Keller-Hunt

- J.M. Keller, D.J. Hunt, *Incorporating Fuzzy Membership Functions into the Perceptron Algorithms*, IEEE Transactions on Pattern Analysis and Machine Intelligence, PAMI-7 (1985), 693–699
- compute fuzzy sets A_1 and A_2
- v_1 and v_2 are clusters centers
- $d_{ki} = d(x_k, v_i)$
- $B = 0.5 \frac{1-\exp(-f)}{\exp(f)-\exp(-f)}$
- $f > 0$ and $m > 1$ are preset constants

Keller-Hunt

1. Chose $c > 0$ and initialize $v \in \mathbb{R}^{p+1}$ arbitrarily and $k = 1$
2. Compute a_{1k} and a_{2k}
   1. If $z_k \in A_1$

$$a_{1k} = 0.5 + 0.5 \frac{\exp\left(f \cdot \frac{d_{k2} - d_{k1}}{d(v_1, v_2)}\right) - \exp(-f)}{\exp(f) - \exp(-f)}$$

$$a_{2k} = 1 - a_{1k}$$

   2. if $z_k \in A_2$

$$a_{2k} = 0.5 + 0.5 \frac{\exp\left(f \cdot \frac{d_{k1} - d_{k2}}{d(v_1, v_2)}\right) - \exp(-f)}{\exp(f) - \exp(-f)}$$

$$a_{1k} = 1 - a_{2k}$$

3. Update v

$$v(new) = \begin{cases} v + c \cdot |a_{1k} - a_{2k}|^m \cdot z_k & \text{if } \overset{a_{1k} \geq 0.5 + B}{a_{1k} \geq 0.5 + B} \text{ and } \overset{\text{or}}{v^T z_k \leq 0} \\ v & \text{otherwise} \end{cases}$$

4. If v has not been modified during last n iterations, then stop, else increase k and goto step 2

(Fuzzy) Relaxation

- separation condition $v^T z \geq b$, with $b \in \mathbb{R}$

1. Chose $c > 0$, $b \in \mathbb{R}$ and initialize $v \in \mathbb{R}^{p+1}$ arbitrarily and $k = 1$
2. Update and normalize v

$$v(new) = \begin{cases} v + c \cdot a_{ik}^2 \cdot \dfrac{b - v^T z_k}{\|z_k\|^2} \cdot z_k & \text{if } a_{ik} > 0.5 \text{ and } v^T z_k \leq b \\ v & \text{otherwise} \end{cases}$$

3. If v is a separation vector, or if the maximum number of iterations has been reached, then stop, else increase k and goto step 2

(Fuzzy) Minimal Square Error

- determine v with $\|v\| = 1$, that minimizes

$$J(v) = \sum_{i=1}^{2} \sum_{j=1}^{n} a_{ij}^2 (v^T z_j - b_j)^2$$

- $b_j > 0$ are preset

$$Z^\star = \left\{ \begin{array}{c} \vdots \\ \sqrt{a_{1k}^2 + a_{2k}^2} z_k^T \\ \vdots \end{array} \right\} \in \mathbb{R}^{n \times p+1}$$

$$b^\star = \left\{ \begin{array}{c} \vdots \\ \sqrt{a_{1k}^2 + a_{2k}^2} b_k \\ \vdots \end{array} \right\} \in \mathbb{R}^{n}$$

$$v = \left(Z^{\star T} Z^\star \right)^{-1} \left(Z^{\star T} b^\star \right)$$

(Fuzzy) Widrow-Hoff

- B. Widrow, M.E. Hoff, *Adaptive Switching Circuits*, 1960 IRE WESCON Convention Record, Part 4, New York, 1960, 96–104

1. Chose $c_i > 0$ $(i = 1, \ldots, n)$, $b \in \mathbb{R}$ and initialize $v \in \mathbb{R}^{p+1}$ arbitrarily and $k = 1$
2. Update v

$$v(new) = v + c_k \cdot (a_{1k}^2 + a_{2k}^2) \cdot (b_k - v^T z_k) \cdot z_k$$

3. If v is a separation vector, or if the maximum number of iterations has been reached, then stop, else increase k and goto step 2

(Fuzzy) Ho-Kashyap

- Y.C. Ho, R.L. Kashyap, *An Algorithm for Linear Inequalities and its Applications*, IEEE Transactions on Elec. Comp., EC-18 (1965), 683–688
- determine v and b that minimize

$$J(v, b) = \sum_{j=1}^{n} (a_{1j}^2 + a_{2j}^2) \cdot (v^T z_j - b_j)^2$$

- compute $Z^\star$ and $b^\star$ as above

$$Z^{\star +} = \left(Z^{\star T} Z^\star \right)^{-1} Z^{\star T}$$

(Fuzzy) Ho-Kashyap

1. Chose $0 < c \leq 1$ and vector b and initialize $k = 1$
2. Compute $b^\star$, $Z^{\star+}$ and set $v = Z^{\star+}b^\star$
3. If $Z^\star v > 0$ then v is a separation vector and stop
4. Else, compute $e^\star$

$$e^\star = Z^\star v - b^\star$$

5. If $e^\star \leq 0$, but not all components are zero, then the fuzzy sets are non linearily separable, and stop
6. If $e^\star \geq 0$, then v is a separation vector and stop
7. Else, update $b^\star$ and v, then increment k and goto step 3

$$b^\star(new) = b^\star + c(e^\star + |e^\star|)$$

$$v(new) = v + cZ^{\star+}\|e^\star\|$$

(Fuzzy) K-Nearest Neighbours

1. Consider set X, fuzzy partition P, constant k and extra point x_0
2. Compute $D(x_0, x_j) = d^2(x_0, x_j)$ for all j
3. Determine permutation σ so that $D(x_0, x_{\sigma(j)})$ is monotonically increasing
4. Compute membership degrees of x_0

$$u_{i0} = \frac{\sum_{j=1}^{k} \frac{u_{i,\sigma(j)}}{D(x_0, x_{\sigma(j)})}}{\sum_{j=1}^{k} \frac{1}{D(x_0, x_{\sigma(j)})}}$$

(Fuzzy) Nearest Prototype

1. Consider set X, fuzzy partition P and extra point x_0
2. Determine prototypes as in FCM
3. Compute membership degree for x_0 with a relation similar to that used in FCM

Restricted Fuzzy C-Means

- H.F. Pop, *Fuzzy Decision Supervised Classifiers*, Studia Universitatis "Babeş-Bolyai", Series Mathematica 40, 3 (1995) 89–100

1. Consider set X, fuzzy partition P and extra point x_0
2. Use FCM on the set $X \cup \{x_0\}$ but do not modify membership degrees u_{ik} for any i and for all $k \neq 0$

Supervised learning versus clustering

- **Clustering**: prototypes and memberships are inferred without preassigned target labels.
- **Supervised classification**: class information is available for training and is used to estimate the decision rule.
- Perceptron-type algorithms learn discriminant boundaries; nearest-neighbour methods classify by local examples.
- Fuzzy class labels or fuzzy training weights affect the learning criterion; they must not be confused with posterior probabilities.

Perceptron and least-squares classifiers

- Perceptron updates respond to misclassified training examples; separability matters for convergence guarantees.
- Pocket algorithms retain a better classifier when the training data are not linearly separable.
- Least-squares and Widrow–Hoff procedures minimize a numerical error criterion, not the number of classification errors directly.
- The choice of objective function determines what an update is optimizing.